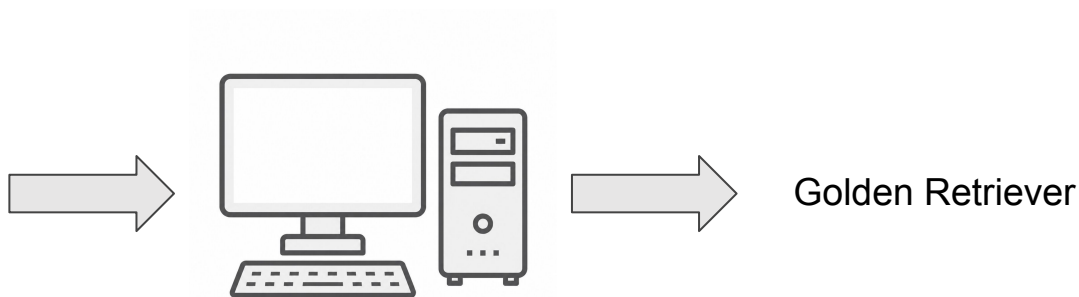


Prototyping Assignment: AI Image Classification System

Gökay Sengün | Lorenz Pusch | 3S | 13. Juli 2026

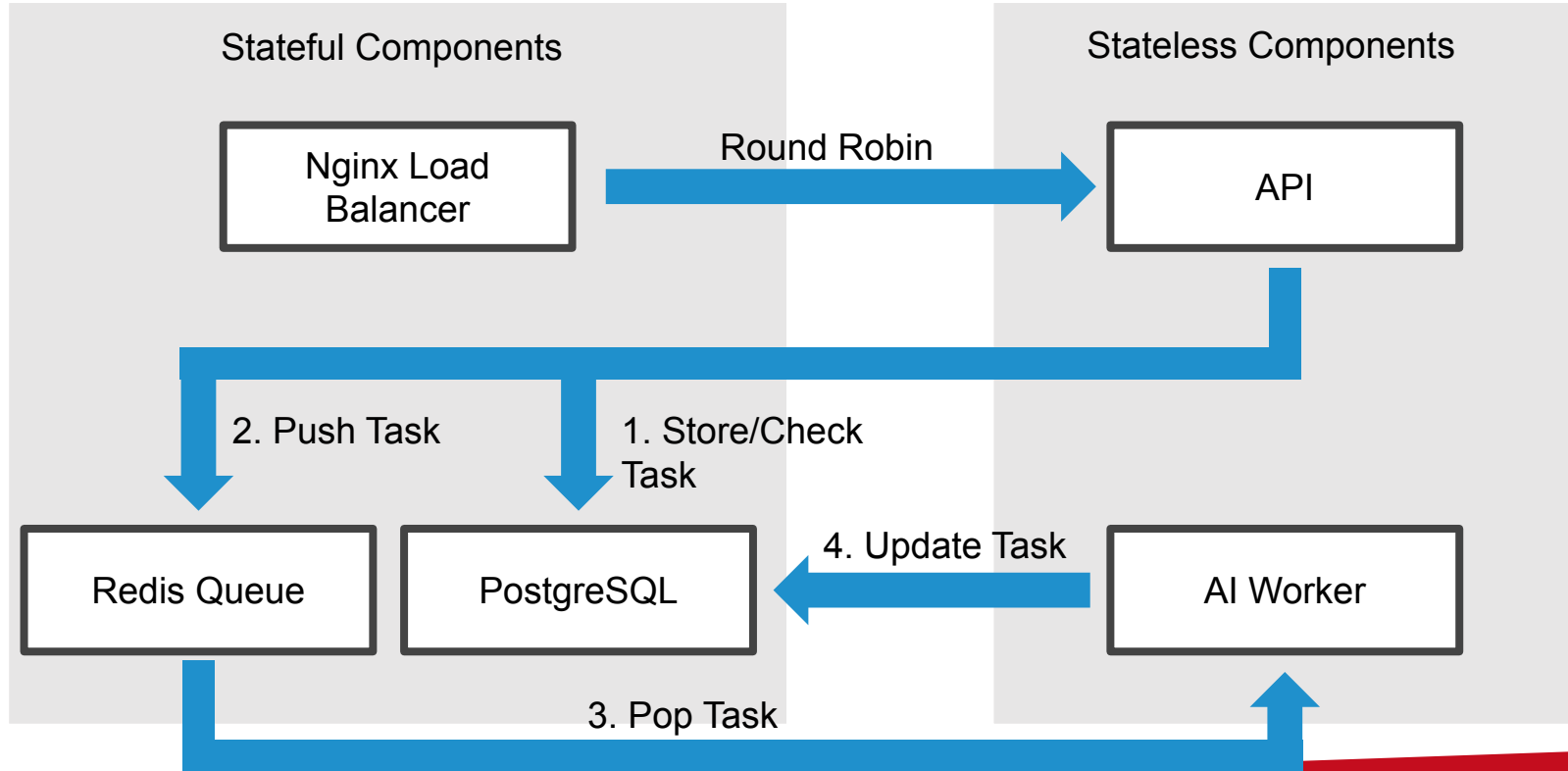


Application

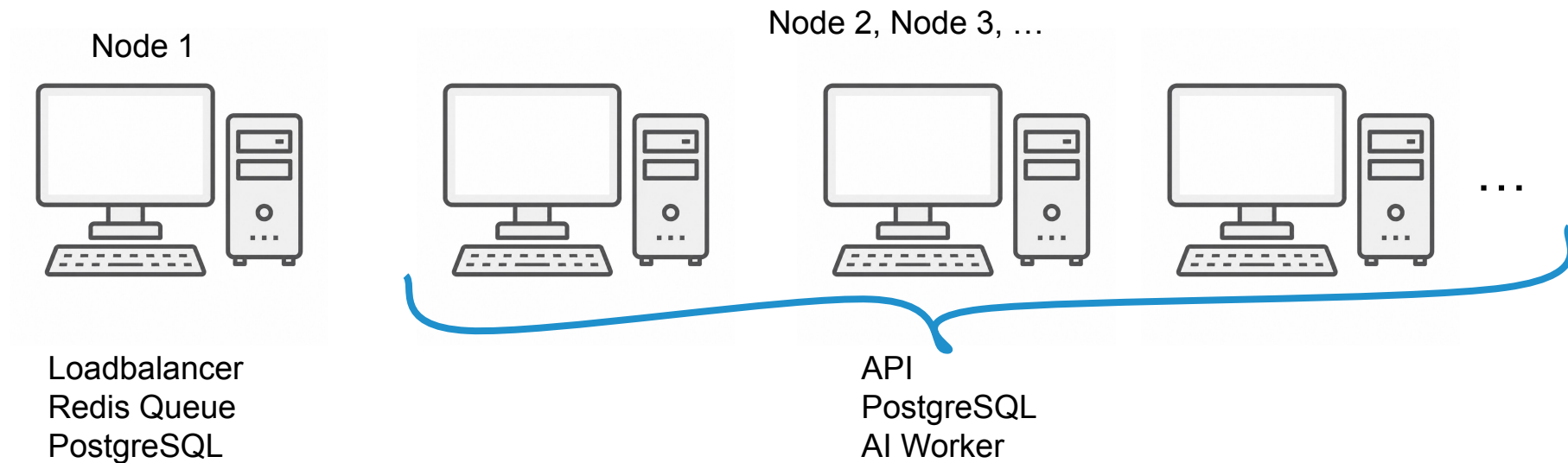


```
gokaysengun@MacBook-Air-von-Gokay ~ % curl -X POST http://34.185.188.250/classify \
-H "Content-Type: application/json" \
-d '{"image_url": "https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcS2FUZMIkoOnHgembMF9InnnlEXenXekksJrA&s", "task_id": "$MY_TASK_ID"}'
{"task_id": "A6683AD3-08CA-4DC6-8CFB-9579FE7E82F2", "message": "Task queued"}%
```

```
gokaysengun@MacBook-Air-von-Gokay ~ % curl http://34.185.188.250/status/$MY_TASK_ID
{"task_id": "A6683AD3-08CA-4DC6-8CFB-9579FE7E82F2", "status": "completed", "result": "golden_retriever", "created_at": "2026-07-05T13:33:55.979112+00:00", "finished_at": "2026-07-05T13:33:56.131783+00:00"}%
```



Horizontal Scaling



Load Shedding & Multi-Tenant Fairness

```
# from app/api/main.py
CLIENT_SOFT_LIMIT          = GLOBAL_MAX_REQUESTS * 0.2
CLIENT_BURST_LIMIT         = GLOBAL_MAX_REQUESTS * 0.5
GLOBAL_THROTTLE_THRESHOLD = GLOBAL_MAX_REQUESTS * 0.8

if client_count >= CLIENT_BURST_LIMIT:
    return Response(content="Client burst quota exceeded.", status_code=429)

if client_count >= CLIENT_SOFT_LIMIT and active_requests >= GLOBAL_THROTTLE_THRESHOLD:
    return Response(content="System under heavy load. Client soft quota exceeded.", status_code=429)
```

Backpressure

According to Little's Law, an insurmountable queue backlog leads to wasted work

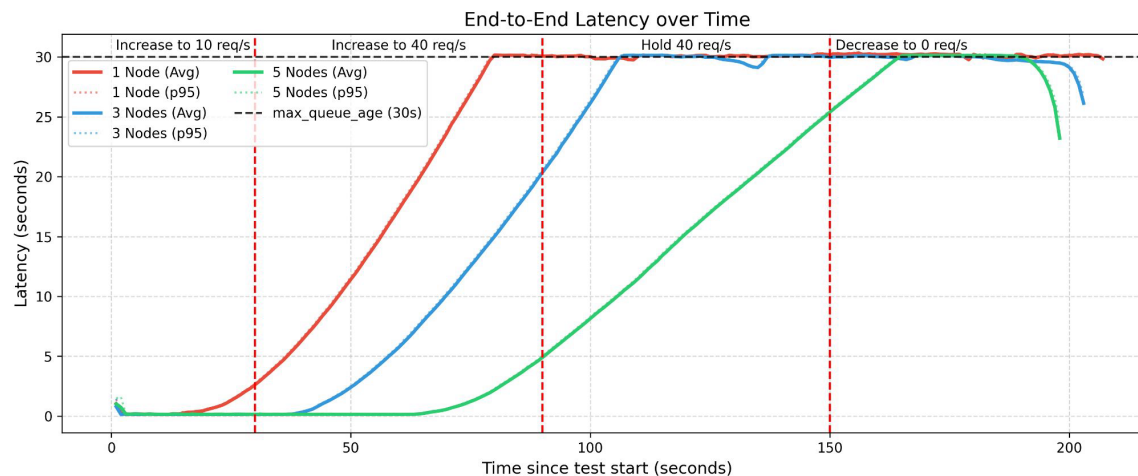
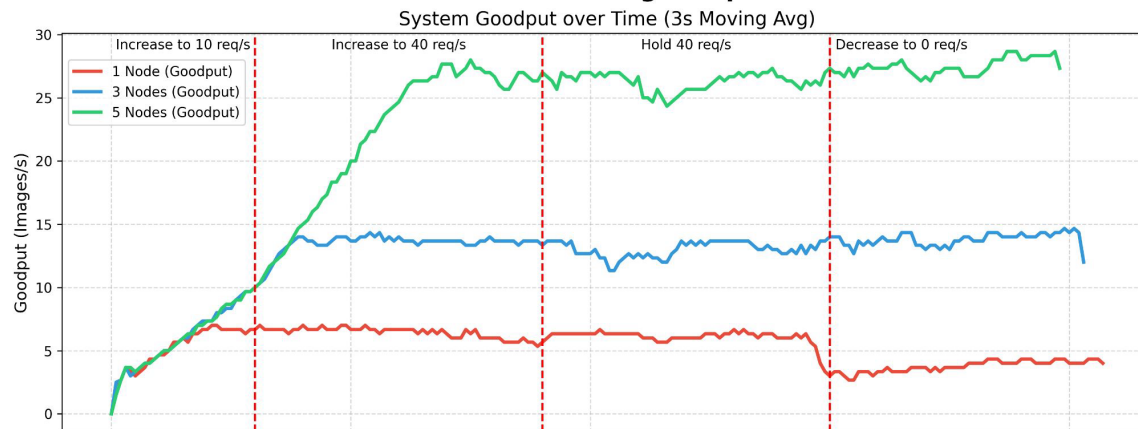
```
# from infra/startup.sh.tpl
MAX_GLOBAL_QUEUE_SIZE=$(( THROUGHPUT * MAX_QUEUE_AGE * NUM_WORKER ))
```

```
# from app/api/main.py
if queue_length >= MAX_QUEUE_SIZE:
    return Response(content="Queue is full. Please retry later.", status_code=503)
```

Results



Horizontal-med Scaling Comparison

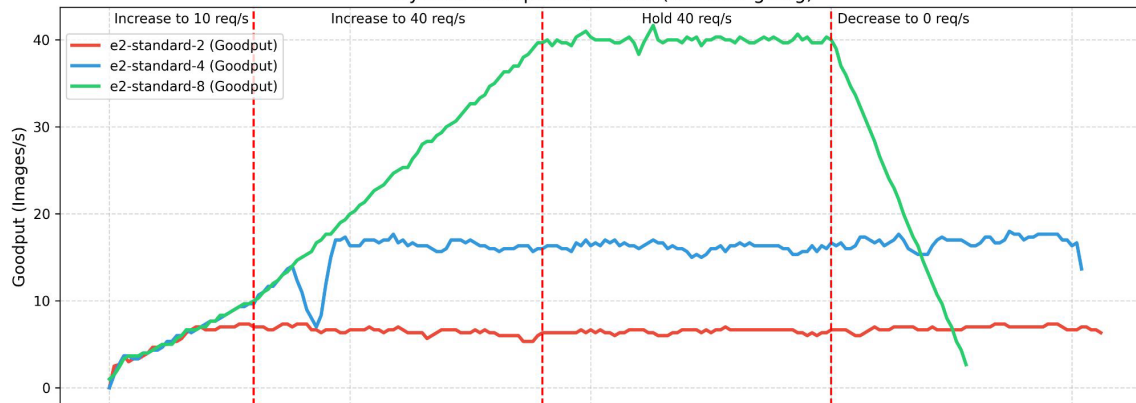


Results

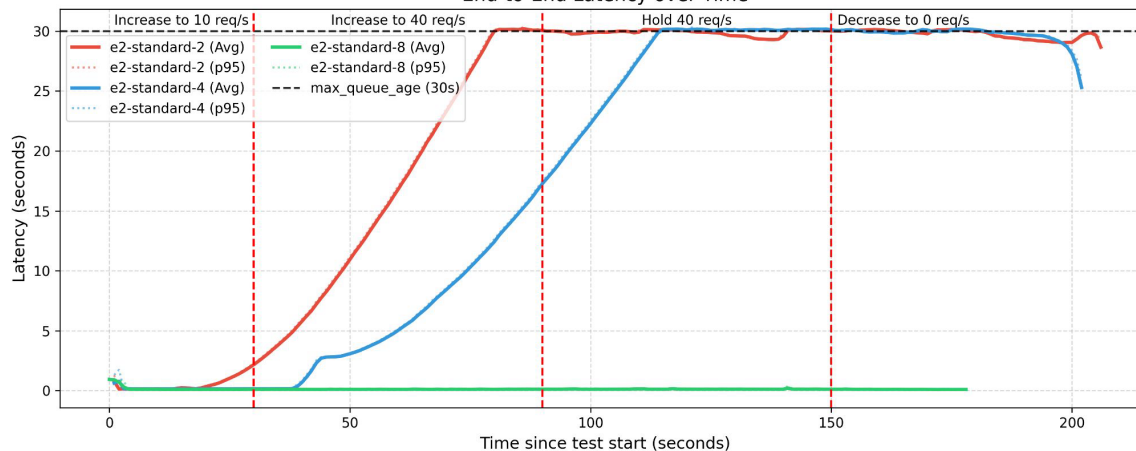


Vertical Scaling Comparison

System Goodput over Time (3s Moving Avg)



End-to-End Latency over Time

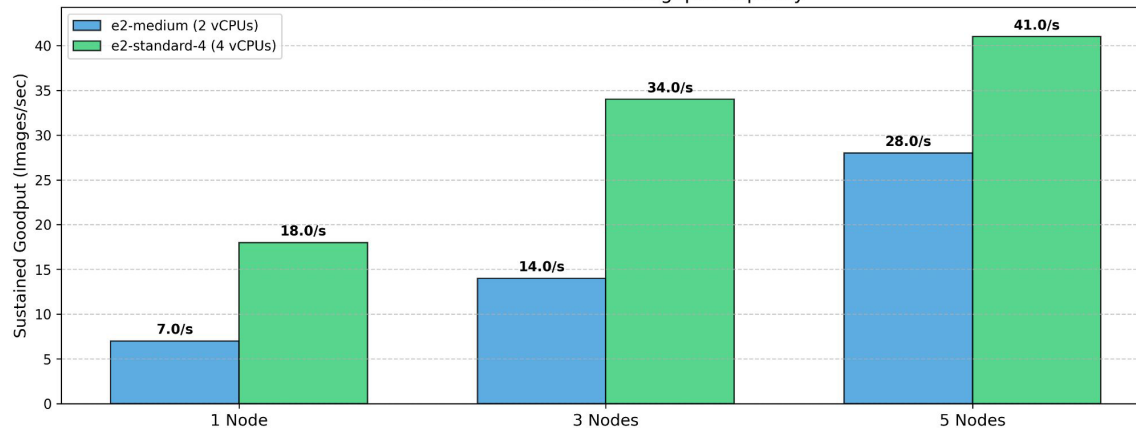


Results

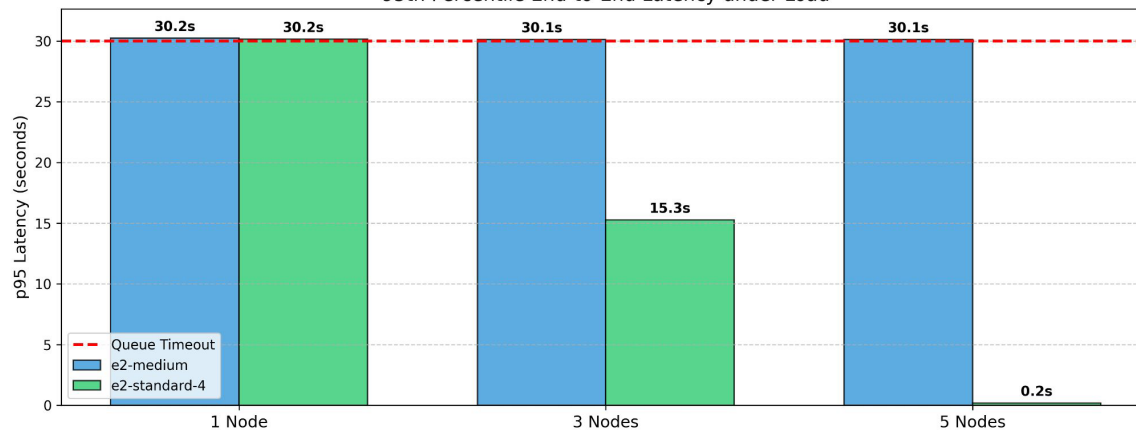


Horizontal & Vertical Scaling Comparison

Maximum Sustained Throughput Capacity



95th Percentile End-to-End Latency under Load



Limitations

- Single Point of Failure (Node 1)
- Dynamic Scaling
- Hyperparameters
 - MAX_API_CAPACITY
 - DB_MAX_CONN
 - THROUGHPUT
 - MAX_QUEUE_AGE

Thank you for listening :)
Questions?